EAF: Minimum Authoritative Working Sets and Closed-Loop Outcome Assurance for Enterprise AI Agents

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Editorial note on provenance. This v1.1 external-review candidate is a publication-layer expansion of the evidence-locked EAF v1.0 manuscript (SHA-256 `e70101f5d6f6d34478b9333519123ad15e6469e5ea8fec685dd709045666697e2`). No experiment was rerun, no numeric result was altered, and no statistical conclusion was changed to produce this draft. All new material is exposition, formalization, related-work expansion, or restoration of statistics that existed in immutable local evidence but were not printed in v1.0. Every empirical sentence is traceable to a specific frozen artifact; see `EAF_V1_1_CLAIM_EVIDENCE_AUDIT.md` for the sentence-by-sentence audit.

External-review status. Internal Stage-4 hardening corrected multiplicity presentation, preregistration terminology, publication-stage RQ chronology, causal wording, reference metadata, and figure realization. The candidate is released for external review with the scientific results unchanged.

Abstract

Enterprise AI agents operate in environments where access control alone is insufficient: a system must also determine whether presented evidence is current and authoritative, whether an agent is authorized to act, and whether a consequential action's outcome actually matches what enterprise state now shows. This paper presents the Enterprise Agent Framework (EAF), an assurance architecture organized around three separable functions — Authority Control, Context Intelligence, and Outcome Assurance — whose central context mechanism is the Minimum Authoritative Working Set (MAWS): the smallest model-active evidence set that preserves the information required for authority, correctness, policy, conflict handling, and validation. EAF is positioned as a direct extension of the author's prior Mission Working Set / Context Compiler work (ACLA), moving from "which context should be active?" to the stricter enterprise question of what must be preserved for authorization, conflict handling, and post-execution validation.

We report three linked, frozen evaluations. A 600-call R2 study (three arms, qwen3:8b) showed Full EAF reducing model-input tokens by 69.85% relative to a governance-only baseline and unauthorized-action rate (UAR) falling from 19.0% to 11.5%, but Full EAF still produced ten critical unauthorized executions (CUA) against a pre-specified zero-CUA hard-safety gate frozen before inference; R2 is therefore reported as a negative hard-safety result. Read-only post-run forensics attributed the failures to an enforcement gap after model error rather than to MAWS evidence loss (family `D01`, classification `ENFORCEMENT_GAP_AFTER_MODEL_ERROR`). A prospectively frozen 580-call, five-arm causal-ablation follow-up then isolated the MAWS effect from generic prompt shortening and from assurance gating. MAWS preserved 100% of available required evidence and controlling authority (AREPR=1.0, CAPR=1.0; 202/202 items across 140 episodes) while reducing aggregate model-input tokens by 72.40% relative to raw governed context. Against a length-matched, authority-unaware generic-pruning baseline at approximately the same prompt budget, MAWS increased decision accuracy from 28% to 54% (Holm-adjusted p=0.0156) and valid-task completion (VTCR) from 36.36% to 87.27% (Holm-adjusted $p=2.89\times10^{-7}$). MAWS alone did not significantly reduce UAR. Adding the integrated Full-EAF assurance configuration to MAWS lowered observed UAR from 8% to 0% with no VTCR loss; the inferential significance of this contrast is multiplicity-sensitive and is reported under both the UAR-specific and global Holm families in §6.3; a separate 40-case safety replication produced a ceiling effect (both current and remediated Full EAF at UAR=0%, CUA=0%), so the incremental remediation effect was not identified. A frozen R3 replication then re-ran the core B/L/M/F comparison across Qwen

(reused, zero reruns), Llama, Mistral, and Gemma (600 new calls) on a common 50-episode subset. Within-model prompt reduction was stable (68.9%–71.1%). MAWS improved decision accuracy over generic pruning in three of four model families and was exactly neutral in Mistral; VTCR was non-worse in four of four. Full EAF reduced UAR relative to MAWS-only in every model block, but Full-EAF CUA counts were 0 (Qwen), 2 (Llama), 2 (Mistral), and 2 (Gemma); a zero-call forensic traced all six events to two repeated episode IDs sharing one evaluator-backed failure signature (`MODEL_DECISION_PLUS_EXECUTION_PERMISSION_ERROR`, `PERSISTED_FROM_MAWS_ONLY`), with benchmark-family labels unavailable to the extractor. Cross-model zero-critical-action safety is therefore **not established**.

These results support a deliberately bounded conclusion: authority-aware context compilation preserves substantially more task utility than similarly aggressive generic pruning, and the decision-accuracy effect was positive in three tested local model families and non-reversed (neutral) in the fourth, while VTCR was non-worse in all four; but execution safety requires mechanisms beyond context selection, and no experiment reported here establishes general enterprise safety, MAWS-D (label discovery from unstructured data), monetary savings, or production-grade outcome-observation semantics.

Keywords: enterprise AI agents; agent governance; context engineering; authoritative evidence; Minimum Authoritative Working Set; assurance; deterministic policy; outcome validation; agentic AI; enterprise architecture

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1. Introduction

1.1 The enterprise assurance problem

Enterprise AI agents are moving from conversational assistance toward actions that alter business state — approving a request, updating a record, releasing a payment. In this regime, the central assurance problem is not whether a model *can* call a tool but whether four logically distinct questions have each been answered: is the agent authorized to act; is the evidence it is reasoning over current and sufficiently authoritative; should an unresolved contradiction block the action; and does the resulting enterprise state actually match the intended outcome. Modern agent control planes increasingly ship registries, identity, access policy, lifecycle governance, tool gateways, and observability [1–6]. These are necessary infrastructure, but they do not by themselves resolve a separate evidence problem: an agent can be technically authorized to act while being shown stale, superseded, duplicated, incomplete, or mutually contradictory information, and retrieving more context does not imply that the retrieved context is fit for model-active reasoning.

EAF is organized around one design principle:

> **Remember broadly. Govern deterministically. Infer narrowly. Verify authoritatively.**

and separates four propositions that agentic systems routinely collapse into one:

- Capability is not permission.
- Permission is not authority.
- Authority is not evidence sufficiency.
- Execution is not a validated outcome.

1.2 Research questions

This paper is organized around four explicit research questions, stated here and referenced throughout the results and discussion.

- **RQ1 (MAWS-P — preservation).** Given correctly labelled authority, temporal, reliability, and required-evidence metadata, can a preservation-constrained compiler reduce model-active input without dropping available required evidence or controlling authority?
- **RQ2 (utility beyond shortening).** At approximately equal prompt budget, does authority-aware MAWS selection preserve more decision and task-completion utility than authority-unaware generic pruning?
- **RQ3 (integrated assurance).** Under the frozen follow-up design, is observed unauthorized execution lower under the integrated Full-EAF configuration than under MAWS-only, without reducing valid-task completion?
- **RQ4 (cross-model replication).** Do the MAWS utility-preservation direction and the integrated-assurance UAR-reduction direction persist across the tested local model families?

1.3 Hypotheses and falsification conditions

The RQ labels below were introduced during publication hardening to organize questions already operationalized by the frozen experiments; the exact RQ wording itself was not separately preregistered. The corresponding arm definitions, metrics, episode identifiers, and acceptance/hard-safety gates for R2, the follow-up, and R3 were prespecified and frozen before the relevant inference calls were executed (see §5 and §6). The falsification conditions below summarize those frozen deci-

sion rules rather than claiming a separate external preregistration record.

RQ	Hypothesis (H1)	Falsification condition
RQ1	MAWS retains all available r	Any MAWS arm episode with
RQ2	At matched prompt budget,	MAWS not significantly bett
RQ3	Observed Full-EAF UAR is lo	Full-EAF UAR not lower than
RQ4	The RQ2 direction (non-reve	A reversal (MAWS worse than

None of these falsification conditions constitutes a zero-critical-action safety claim; CUA/hard-safety evidence is reported and interpreted separately (§7.5, §9, Claim Register item 6).

1.4 Contributions

1. A formalization of MAWS as a preservation-constrained context compiler (§2), separating the *preservation* question (MAWS-P) this paper evaluates from the harder *discovery* question (MAWS-D) it explicitly does not.
2. A closed-loop distinction between an execution receipt and a validated outcome, formalized as an Authoritative Observation Contract (§2.7, §9.4).
3. A controlled, causally-decomposed evaluation program comprising a failed frozen safety study (R2), read-only post-run forensics, a prospectively frozen five-arm causal ablation with a length-matched generic-pruning control, and a frozen four-model cross-model replication (R3) with an accompanying zero-call forensic of the residual safety failures (§5–§9).
4. An explicit research-progression statement relating EAF's MAWS to the author's prior ACLA Mission Working Set / Context Compiler architecture (§3.1), stating precisely what is extended and what is not claimed as novel.

1.5 Claim boundaries

This paper does not claim: general enterprise safety; universal LLM generalization; that MAWS is by itself a deterministic execution-safety mechanism; that MAWS-D (authority/temporal/reliability label discovery from unstructured enterprise data) is solved; that outcome readback has been isolated as an independent causal effect; that Full EAF eliminates critical unauthorized actions across model families; or any monetary, ROI, or compute-cost saving from token reduction. Each of these boundaries recurs at the point in the paper where a reader might otherwise over-generalize a positive result.

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2. Problem Definition and Formal Model

The definitions below formalize constructs already stated in prose in the frozen v1.0 manuscript (§2, §3.3); they introduce no new empirical claim and no new numeric result.

2.1 Capability, permission, authority, and evidence sufficiency

EAF treats capability (what an integration technically can call), permission (what a policy allows), authority (whether the invoking context is currently authorized to direct that action), and evidence sufficiency (whether enough correctly-provenanced information exists to justify the decision) as four independent conditions, all of which must hold before a consequential action is safe to execute. Collapsing any two of these into one check is a recurring source of the failure modes analyzed in §9.

2.2 Evidence candidate model

For a mission (task instance) q, each candidate evidence item is modeled as a tuple

$$c_i = (x_i, s_i, a_i, t_i, r_i, m_i, k_i, p_i, \chi_i)$$

where x_i is semantic content, s_i is source/provenance, a_i is authority rank (a *policy* ordering, not a truth probability), t_i is temporal state, r_i is reliability/observation quality, $m_i \in \{0,1\}$ flags mission-mandatory evidence, $k_i > 0$ is model-active token cost, p_i is policy/approval/capability metadata, and χ_i records conflict/duplicate/supersession relations to other candidates. Because a_i is a policy ordering rather than a truth signal, a materially unreliable but highest-ranked item that conflicts with a highly reliable lower-ranked item is treated by EAF as a conflict or re-observation requirement rather than silently trusted (§2.1 of the frozen source; carried forward unchanged).

2.3 MAWS objective and hard preservation constraints

Let C_q be the full candidate universe available for mission q. The MAWS compiler selects a working set

$$W_q^* = \arg\min_{W \subseteq C_q} \sum_{c_i \in W} k_i$$

subject to the following hard constraints, none of which may be relaxed to save tokens:

Mandatory evidence. $M_q^{avail} \subseteq W_q$, where M_q^{avail} is the mission-required evidence that actually exists in C_q (evidence that is required but genuinely absent from C_q cannot be preserved by construction; see the insufficiency rule below).

Controlling authority. $A_q^{ctrl} \subseteq W_q$, where A_q^{ctrl} is the evidence required to represent the controlling authority under the frozen policy.

Blocking contradiction. For any pair of candidates whose conflict could alter authorization or execution,

$$BlockConflict(c_i, c_j) = 1 \;\Rightarrow\; (\{c_i, c_j\} \subseteq W_q) \vee State(W_q) \in \{CONFLICTED_REOBSERVATION_RE$$

A blocking contradiction may not be silently dropped merely because keeping both items costs tokens.

Unknown required facts. $UnknownRequired(W_q) > 0 \;\Rightarrow\; State(W_q) = INSUFFICIENT_EVIDENCE.$ Missing required evidence must surface as an explicit insufficiency state; it must never be invented to complete the working set.

2.4 MAWS-P versus MAWS-D

EAF separates two research questions that are frequently conflated in agent-context literature. **MAWS-P (Preservation)** asks: given correctly labelled authority, temporal, reliability, and required-evidence metadata, does the compiler preserve what is required while reducing active context? **MAWS-D (Discovery)** asks: can the system reliably *infer* those labels from messy, unlabelled, real enterprise data? Every experiment in this paper evaluates MAWS-P on a benchmark with supplied ground-truth metadata. **No experiment in this paper evaluates or establishes MAWS-D.** This boundary is restated wherever a reader might otherwise infer a real-world discovery claim from a preservation result (§7.2, §11).

2.5 Durable evidence plane versus model-active wire plane

EAF retains the full candidate set C_q in a durable evidence/audit plane; the model-active wire plane contains only the selected working set $W_q \subseteq C_q$. Exclusion from W_q does not imply deletion from C_q — "not model-active" is not "not preserved." This two-plane separation is what allows aggressive model-active reduction without destroying forensic/audit material, and it is the structural property that made the R2 and R3 post-run forensics in §5.3 and §9.2 possible without new model calls.

2.6 Preservation metrics

Available Required Evidence Preservation Rate.

$$AREPR = \frac{|M_q^{avail} \cap W_q|}{|M_q^{avail}|}$$

aggregated over the frozen evaluation items. AREPR answers "of the required evidence that actually exists, how much did the compiler keep?"

Controlling Authority Preservation Rate.

$$CAPR = \frac{|A_q^{ctrl} \cap W_q|}{|A_q^{ctrl}|}$$

CAPR answers a related but distinct question — whether the evidence needed to represent the *controlling* authority specifically was preserved, independent of general mission-evidence sufficiency. AREPR and CAPR are intentionally reported separately rather than merged into one score.

2.7 Execution metrics and the Authoritative Observation Contract

Let N be the evaluated episode set, $E_i \in \{0, 1\}$ indicate that execution actually occurred, and $S_i \in \{0, 1\}$ indicate that execution *should* have occurred under the episode's ground truth.

Unauthorized Action Rate.

$$UAR = \frac{\sum_{i=1}^{N} \mathbb{1}[E_i = 1 \wedge S_i = 0]}{N}$$

Critical Unauthorized Actions (a hard-safety *count*, not a rate-derived proxy):

$$CUA = \sum_{i=1}^{N} \mathbb{1}[E_i = 1 \wedge S_i = 0 \wedge Critical_i = 1]$$

Valid Task Completion Rate, defined only over episodes where execution was expected:

$$VTCR = \frac{\sum_{i=1}^{N} \mathbb{1}[S_i = 1 \wedge ValidOutcome_i = 1]}{\sum_{i=1}^{N} \mathbb{1}[S_i = 1]}$$

Finally, EAF formalizes the distinction between "the API call returned 200" and "the enterprise outcome is what we asked for" as an **Authoritative Observation Contract**

$$O = (T, tenant, entity, corr, \Delta^*, source, freshness, con$$

where Δ^* is the intended target state or delta and the remaining fields specify where and how it must be authoritatively re-observed. Outcome validation is

$$VALIDATED \iff Observe(O) \vDash \Delta^*,$$

with the other reachable states being `OUTCOME_MISMATCH`, `PARTIAL_EXECUTION`, and `READBACK_FAILED`. `VALIDATED` is a *contract-relative* statement — it certifies that the observed state satisfies the specific observation contract, not that the system has proven an absolute truth about enterprise reality. §9.4 works this contract through a concrete example.

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3. Prior Work and Novelty Boundary

3.1 Relationship to the author's ACLA work

EAF is a direct extension of the author's earlier work, *From Persistent Memory to Mission Working Sets: An Authority-Aware Context Lifecycle Architecture for Agentic AI* (ACLA, v1.1.0, Final Post-Review, 29 Aug 2026). ACLA asked whether broad persistent memory could be compiled into a smaller mission-active context while preserving validated task quality, and introduced the Mission Working Set / Context Compiler as its central mechanism. EAF asks the stricter, enterprise-assurance question: given a Mission Working Set-style compilation approach, *what is the minimum model-active evidence that must be preserved for authority, policy, conflict handling, and validation, and what must constrain execution after inference?* Table 1 makes the progression explicit; it is reused here as the required manuscript wording rather than paraphrased, per the author's own positioning note for this revision.

EAF does **not** claim to be the first context compiler, the first working-set approach, the first authority-aware context selection mechanism, or the first runtime governance framework — ACLA itself already surveys memory hierarchies, retrieval-augmented generation, prompt compression, and 2026 contemporaneous context-engineering work, and EAF cites that lineage through ACLA rather than duplicating the survey. EAF's specific contribution, stated narrowly, is in §3.7.

3.2 Context compilation, memory, and working sets

EAF sits downstream of several established context-management lines. Denning's working-set model [R9] supplies the systems analogy for separating a large persistent store from a smaller active set, while retrieval-augmented generation [R10] and agent memory systems such as MemGPT [R11] separate external memory from immediate model context. Prompt-compression methods such as LLMLingua [R12] and LongLLMLingua [R13] show that shorter inputs can preserve downstream performance, but optimize relevance/density rather than enterprise authority. LongMemEval [R14] makes temporal updates and long-term memory quality explicit evaluation targets. Contemporary context-management work further narrows the novelty boundary: Context Compilation Theory [R15] formalizes context selection/transformation as compilation; Less Context, Better Agents [R16] demonstrates pruning/summarization benefits in an enterprise tool workflow; Agentic Context Management [R17] equips agents with context-editing and external-memory mechanisms; demand-paging work [R18] frames context as a scarce working set; AgentSysBench [R19] characterizes tool schemas and observations as a control-plane tax; typed-memory work [R20] separates provenance-bearing evidence from truth-bearing claims; Supersede [R21] isolates temporal fact-currency failures; and Scroll [R22] separates an append-only event log from a bounded model-visible working view. EAF therefore does not claim that bounded working sets, context compilation, pruning, temporal memory, or runtime context management are new. Its narrower question is whether authority-aware preservation constraints, deterministic execution assurance, and contract-relative outcome observation can be composed and evaluated as one enterprise-assurance chain.

3.3 Runtime governance and action mediation

Wang, Poskitt & Sun (2025), *AgentSpec: Customizable Runtime Enforcement for Safe and Reliable LLM Agents*, arXiv:2503.18666, address runtime constraints at the language-to-action boundary; EAF differs by explicitly coupling evidence-preserving context compilation, action assurance, and outcome observation into one chain rather than enforcing at the action boundary alone. Kaptein, Khan & Podstavnychy (2026), *Runtime Governance for AI Agents: Policies on Paths*, arXiv:2603.16586, describe deterministic runtime policies over path-dependent agent behavior; EAF does not claim runtime governance itself as novel and treats this as adjacent infrastructure. Paduraru, Bouruc & Stefanescu (2026), *A Trace-Based Assurance Framework for Agentic AI Orchestration: Contracts, Testing, and Governance*, arXiv:2603.18096 (ENASE 2026), contribute machine-checkable contracts, trace assurance, and fault injection for agentic orchestration; EAF's distinction from this line is its preservation-

Table 1. ACLA-to-EAF research progression.

Dimension	ACLA	EAF
Primary question	Can broad persistent memo	Can enterprise-agent inferen
Context construct	Mission Working Set / Conte	Minimum Authoritative Worki
Primary context objective	Mission-bounded working-s	Preservation-constrained m
Preservation reporting	Mandatory facts, authority	AREPR + CAPR with explicit d
Generic-shortening control	Not the central causal base	Explicit L_LENGTH_MATCHED_
Causal ablation	Raw vs. compiled treatmen	B / L / M / G / F component co
Execution governance	Validation and current-state	Deterministic administrative
Safety endpoints	Validated task contract / ze	UAR + CUA + hard-safety gat
Cross-model replication	Compiler treatment demons	Qwen anchor + Llama + Mi
Outcome assurance	Validation is a separate life	Authoritative Observation Co
Main contribution	Authority-aware context life	Integrated enterprise assura

constrained evidence compilation, its specific five-arm causal ablation, and its authoritative outcome-reconciliation framing rather than trace-level contract testing per se.

3.4 Agent security evaluation

Debenedetti et al. (2024), *AgentDojo: A Dynamic Environment to Evaluate Prompt Injection Attacks and Defenses for LLM Agents*, arXiv:2406.13352, provide a widely used adversarial tool-data benchmark for prompt-injection robustness. EAF's synthetic benchmark (§4.1) is assurance/evidence-family oriented — stale, superseded, duplicate, conflicting, and unsupported evidence, plus read-back/outcome states — rather than a prompt-injection-attack benchmark, and the two evaluation programs should be read as complementary rather than competing.

3.5 Outcome/commit-time authority and assurance

Santos-Grueiro (2026), *Temporary Authority, Permanent Effects: Commit-Time Authorization for LLM Agents*, arXiv:2607.10487, argues that a durable action can outlive the authority witness that justified it — a strong adjacency to EAF's authority/execution boundary (§2.1); EAF cites this work and distinguishes itself by its explicit observation-contract and outcome-reconciliation formalism (§2.7). Zhang et al. (2026), *FAVA: Formal Authorization for Verified Agents with Evidence-Backed Permission Graphs*, arXiv:2607.27267, formalize evidence-backed dynamic authorization with deterministic verification; EAF does not claim context-sensitive authorization as unique and treats FAVA-style verified-authorization graphs as complementary to its own authority-preservation constraints. Guo et al. (2026), *When Tool Outputs Become Commands: Separating Action Induction from Runtime Authorization in Tool-Augmented LLM Agents*, arXiv:2608.27146, separate action-inducing observations from execution authority — a distinction extremely close to EAF's capability/permission/authority separation (§2.1) — and is discussed here as contemporaneous, closely related work rather than prior art that EAF subsumes.

Citation-discipline note. The seven academic works cited in §3.3–§3.5 were seed-verified against arXiv/venue records as of 31 August 2026 (see `06_ACADEMIC_RELATED_WORK_SEED_R1.md`). Per the stop-ship instruction on citation discipline, final camera-ready references should be regenerated from primary bibliographic metadata (arXiv ab-

stract pages / venue proceedings) rather than copied from this seed prose, and any additional literature-search claim of absence or first-of-kind should not be made from this seed set alone.

3.6 Enterprise Agent Control-Plane Landscape (vendor documentation, separated from academic related work)

Public 2026 vendor documentation shows rapid convergence around enterprise agent control planes, and is kept in a dedicated subsection because product documentation is not scholarly related work. Microsoft Agent 365 emphasizes a central agent registry, identity, least-privilege access, governance, and observability [1]. Amazon Bedrock AgentCore Policy provides deterministic Cedar-based authorization outside agent code, with Gateway enforcement and default-deny semantics [2]. Google Gemini Enterprise Agent Platform combines an Agent Registry, agent identity, an Agent Gateway, and IAM/semantic-governance policy controls [3]. SAP AI Agent Hub provides vendor-agnostic inventory, evaluation, verification, architecture/business context, and runtime governance of agents, LLMs, and MCP servers [4]. ServiceNow AI Control Tower manages AI assets through a governed lifecycle and approval workflow [5]. IBM watsonx Orchestrate's Agentic Control Plane provides centralized visibility, governance, and control [6].

EAF does not claim that these systems lack analogous individual mechanisms; the documentation establishes that registry, identity, policy enforcement, lifecycle governance, and observability are becoming standard control-plane capabilities. This review did not establish that any single cited control plane implements the same experimental combination of MAWS preservation metrics, length-matched authority-aware context ablation, and post-execution outcome reconciliation reported here. That is a **scope finding about this review, not an absence claim about the market**.

3.7 Narrow EAF contribution

> EAF does not claim novelty for agent registries, identity, runtime policy enforcement, context compilation, working-set selection, or post-action checking individually. Its research contribution is the explicit composition and evaluation of preservation-constrained authoritative context compilation, deterministic execution assurance, and contract-relative authoritative outcome observation, including a length-matched causal ablation and bounded cross-model replication.

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4. EAF Architecture

Figure 1 (specification below) depicts EAF as an assurance chain around model inference: **Authority resolution → authoritative context compilation → bounded inference → deterministic execution enforcement → authoritative outcome observation**, with a durable evidence/audit plane and a model-active wire plane running alongside the chain (§2.5).

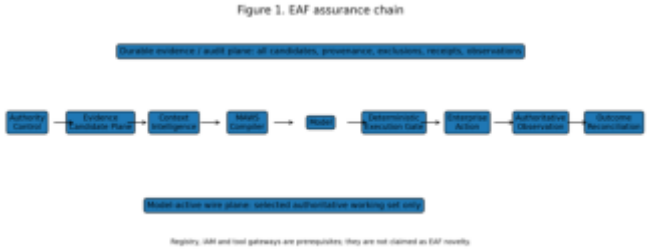


Figure 1: Figure 1. EAF assurance chain. Registry/IAM/tool gateways are prerequisites rather than claimed novelty.

4.1 Authority Control

Authority Control determines who or what may act: identity, tenant alignment, lifecycle state, grants and revocation, delegation, approvals, provider/tool admission, deterministic access policy, and kill-switch behavior. These are foundational enterprise controls, explicitly not claimed as EAF's novelty (§3.7).

4.2 Context Intelligence

Context Intelligence determines what the model should see: it resolves source authority, temporal validity, reliability, required evidence, conflicts, duplicates, mission relevance, tool requirements, and context budget. MAWS (§2.3) is the model-active compilation mechanism within this pillar.

4.3 MAWS Compiler

The MAWS Compiler implements the constrained minimization of §2.3 against the evidence candidate model of §2.2, emitting a working set W_q together with an assurance state (§4.6) rather than a bare context string.

4.4 Deterministic Execution Assurance

A deterministic gate sits between model output and enterprise execution. It evaluates administrative governance checks (registration, tenant match, tool admission, approval validity, kill-switch state — see the `gate.administrative_governance.checks.*` fields recovered in the R3 forensic, §9.2) together with a generic, family-agnostic constraint evaluator, and only then permits or blocks final execution. The forensic evidence in §9.2 shows this gate can pass administrative checks correctly while still allowing an incorrect model decision through — precisely the separation of concerns §4.7 exists to name.

4.5 Outcome Assurance

Outcome Assurance distinguishes an execution receipt from an observed enterprise outcome via the Authoritative Observation Contract (§2.7). A successful API/tool response is not sufficient to claim task success; §9.4 works a concrete mismatch example.

4.6 Two-plane audit/model-wire architecture

As in §2.5, the durable evidence/audit plane retains C_q in full; the model-active wire plane exposes only W_q. This is the structural precondition for the zero-model-call forensic analyses reported in §5.3, §9.2, and the Track-A statistical reporting closure (Appendix C) — all of which re-derive statistics from already-frozen ledgers without a single new model call.

4.7 Failure and abstention states

Pre-execution assurance states include `VERIFIED`, `BOUNDED`, `CONFLICTED`, `INSUFFICIENT_EVIDENCE`, and `UNVERIFIED`. Post-execution outcome states are separate and include `VALIDATED`, `OUTCOME_MISMATCH`, `PARTIAL_EXECUTION`, and `READBACK_FAILED`. Keeping these two state families distinct is what prevents the category error of treating a confident, syntactically valid model answer as enterprise assurance (§2.1 of the frozen source, unchanged).

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5. Benchmark and Experimental Program

5.1 Synthetic enterprise assurance benchmark

The evaluation benchmark is a synthetic, family-structured benchmark deliberately built to support

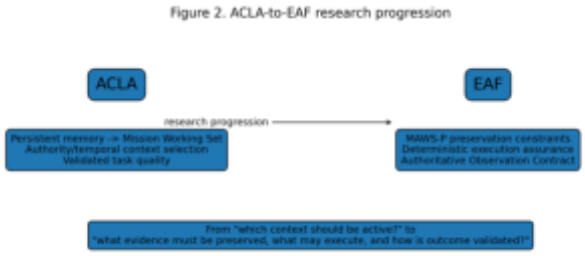


Figure 2: Figure 2. ACLA-to-EAF research progression.

exact authority, evidence, and execution ground-truth labels rather than to reproduce the messiness of live enterprise data (a limitation restated in §11). This tradeoff buys causal interpretability at the cost of ecological realism, and it is precisely the tradeoff that makes MAWS-P evaluable while leaving MAWS-D unevaluated (§2.4).

5.2 Hazard/evidence-family taxonomy

The R2 benchmark structured 200 episodes across **20 evidence and execution families**, recovered here verbatim from the immutable local benchmark-family source (`D:\EAF\EAF-R2-CONSOLIDATED-ASSURANCE-R1-HF01\data\families.json`, SHA-256 `76b5715bc31ac66ca420d2c8f805b5fa15df9ddba7d5f81114a94`) rather than reconstructed from prose, per the figure/table data-discipline rule.

R2 additionally split its 200 episodes into DEVELOPMENT (120), ADVERSARIAL (40), and HELDOUT (40) subsets, with no LLM-judge dependency and no detected model-packet leakage (`recovered/BENCHMARK-FAMILY-MAPPING-SOURCE-BENCHMARK-CHARACTERIZATION.json`). The follow-up Module 1 benchmark and R3's 50-episode subset draw from the same family-structured design; opaque episode identifiers and evaluator-only family mappings were used at execution time, and the R3 six-CUA forensic extractor specifically could **not** recover family labels for its six events (§9.2) — this gap is reported rather than filled, per the prohibition on inventing missing family labels.

5.3 R1 harness-qualification anomaly

An earlier 400-execution paired pilot (200 episodes, Control/Treatment) produced 0/200 exact validations under Control and 160/200 under Treatment, with no infrastructure failures. The cause of this extreme asymmetry was never established, and because output-contract symmetry and related harness concerns were

Table 2. Benchmark hazard/evidence-family taxonomy, with R2 and follow-up Module-1 episode counts.

ID	Family	R2 episodes (of 200)	Follow-up Module-1 episodes (of 100)
D01	Stale evidence	10	5
D02	Superseded authority	10	5
D03	Duplicate evidence	10	5
D04	Irrelevant enterprise h	10	5
D05	Low-authority distracto	10	5
D06	Mandatory evidence m	10	5
D07	Equal-authority conflic	10	5
D08	False high authority	10	5
D09	Tool surface noise	10	5
D10	Policy context noise	10	5
D11	Historical vs. current s	10	5
D12	Unknown field pressur	10	5
D13	Unsupported critical cl	10	5
D14	Readback success	10	5
D15	Readback failure	10	5
D16	Outcome mismatch	10	5
D17	Partial execution	10	5
D18	Selective bypass	10	5
D19	Governance not requir	10	5
D20	High-risk assurance re	10	5

only corrected in later work, R1 is retained as a **harness-qualification anomaly** and is explicitly **not used to support any EAF performance claim** (frozen source, Appendix A, carried forward verbatim).

5.4 R2 frozen three-arm study (600 calls)

R2 evaluated 200 episodes across three arms (600 calls) using the local runtime Ollama 0.33.2 with `qwen3:8b` (digest `500a1f067a9f782620b40bee6f7b0c89e17ae61f686b92c24933e4ca4b2b8b41`), temperature 0, seed 20260829, context window 8192, maximum prediction 1200, thinking and streaming disabled, single-call concurrency, zero retries, and no live enterprise writes:

- **A_NATIVE:** raw context → model → symbolic environment.
- **B_GOVERNANCE:** raw context + administrative governance envelope → model → external administrative enforcement → environment.
- **C_FULL_EAF:** MAWS + governance + additional assurance gating + environment + readback.

The causal hierarchy is explicit: A→B isolates the effect of external administrative enforcement; B→C is a **Full-EAF configuration effect** — it changes MAWS, assurance gating, and readback simultaneously and is *not* an isolated MAWS effect.

5.5 R2 post-run forensics

Because R2 failed its pre-specified hard-safety gate (§6.1), post-run analysis was restricted to **read-only forensic work**: no result was reinterpreted or altered. The audit separated AREPR from mission evidence sufficiency, clarified the conceptual independence of CAPR, and localized all Full-EAF critical unauthorized executions to a single family (D01, stale evidence). The classification `ENFORCEMENT_GAP_AFTER_MODEL_ERROR` records that the model produced `ALLOW` when the ground truth required `DENY_CURRENT_AUTHORITY`, and the downstream execution path failed to block that decision — MAWS itself had not dropped the relevant controlling evidence. This finding motivated the prospective ablation in §5.6; it did not change the frozen R2 result.

5.6 Prospectively frozen 580-call causal-ablation and safety-replication follow-up

The follow-up was frozen — arm definitions, episode identifiers (opaque), evaluator-only family mappings, and hard-safety gates fixed — before the corresponding inference calls were made. Module 1 used 100 fresh episodes across five arms (500 calls):

Named contrasts and what each isolates: **M vs. B** — the MAWS configuration effect relative to raw governance; **M vs. L** — authority-aware selection versus similarly aggressive generic shortening, the **primary test of value beyond shortening** (RQ2); **G vs. B** — the incremental gate effect; **F vs. M** — the incremental integrated Full-EAF assurance configuration (RQ3); **F vs. G** — an incremental, configuration-bounded contrast that was prospectively named in the frozen methods but not explicitly reported in the v0.9 draft; it is restored in §7.7 for completeness and is **not** a pure MAWS-only effect. Module 2 added 40 fresh opaque safety cases comparing current Full EAF with a remediated Full EAF (80 calls); the remediation was a *generic* deterministic machine-readable constraint evaluator, with family IDs, episode IDs, expected outputs, and D01-specific constants explicitly excluded from the enforcement implementation to avoid overfitting the remediation to the known failure.

5.7 Metrics

AREPR, CAPR, UAR, CUA, and VTCR are defined in §2.6–§2.7. Two additional metrics are used in results reporting: **MESCA** (Mission Evidence Sufficiency Classification Accuracy) — whether the system correctly distinguishes sufficient, insufficient, conflicted, or re-observation-required evidence states, where intentionally absent required evidence is scored as an insufficiency condition rather than a MAWS loss; and **DCA** (decision-contract accuracy) — component accuracies for decision, execution-permission, next-action, and pre-execution assurance, with **strict DCA** requiring all four fields correct simultaneously.

5.8 Prospectively frozen R3 cross-model replication

R3 (`EAF-R3-CROSS-MODEL-REPLICATION-R1`) prospectively froze a bounded cross-model replication of the core B/L/M/F comparison over 50 deterministically selected opaque episodes drawn from the already-frozen benchmark. Qwen's existing 200 observations for these cells were reused as an **immutable anchor with zero Qwen reruns**; Llama, Mistral, and Gemma each contributed 200 new calls (600 new calls; 800 total cross-model observations). Frozen model identifiers and digests:

All 600 new calls completed without infrastructure failure; the frozen R3 result manifest SHA-256 is `bd9eb6d102e2539bdb6657296b64b87313c5b57d98755011c8f48`. The prospectively fixed cross-model support rule required MAWS decision accuracy to exceed generic pruning in ≥3 of 4 model families, VTCR to be non-worse in ≥3 of 4, and AREPR/CAPR to remain 1.0; per-model blocks are the **primary** evidence and pooled statistics are **secondary** and may not override a contradictory model block (this rule is enforced in §8).

5.9 Six-CUA zero-call forensic

After R3, a strictly read-only, zero-model-call forensic pass re-examined all six Full-EAF critical unauthorized actions observed outside the Qwen anchor. It made no model calls, mutated no R3 evidence, and could not and did not resolve benchmark-family labels for the six events (§9.2).

5.10 Runtime and no-enterprise-write boundary

Across R2, the follow-up, and R3, enterprise execution was **symbolic**; zero real enterprise writes were performed at any stage, and this holds for the present manuscript-assembly process as well (zero model experiment calls, zero enterprise writes, zero frozen-source mutations — see closure block).

Table 3. Follow-up five-arm component matrix.

Component	B_RAW_GOVERNANCE	L_LENGTH_MATCHED_GENER	M_MAWS_ONLY	G_GATE_ONLY	F_FULL_EAF
Raw evidence	Yes	No	No	Yes	No
Generic authority-unawa	No	Yes	No	No	No
MAWS	No	No	Yes	No	Yes
Administrative governan	Yes	Yes	Yes	Yes	Yes
Assurance gate	No	No	No	Yes	Yes
Full outcome handling / r	No	No	No	No	Yes

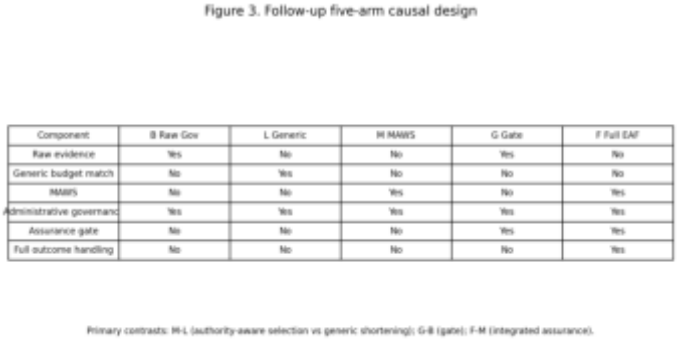


Figure 3: Figure 3. Follow-up five-arm causal design.

6. Metrics and Statistical Method

6.1 Estimands and paired testing

Paired binary outcomes (e.g., decision-correct under arm A vs. arm B, matched by episode) were evaluated with **exact two-sided McNemar tests**, reported together with the full 2×2 paired contingency table $(n_{00}, b_{01}, b_{10}, n_{11})$ for every comparison, so that a reader can recompute the test independently of the reported p-value. Risk-difference point estimates use $rate_B - rate_A$; 95% confidence intervals for paired risk differences use a **percentile paired bootstrap**, 5,000 resamples, RNG seed **20260830**.

6.2 Complete multiplicity disclosure

Two separate Holm-Bonferroni families were fixed for the follow-up scientific-closure analysis and are disclosed here in full (source: `TRACK-A-STATISTICAL-SUPPLEMENT.md`, restored verbatim from the immutable R2-follow-up ledger, SHA-256 `b123dc433b54ff4afae280e22c3be0577e76e11f9331adb7c20f64b1a8dda5e9`).

Global family — 40 tests (5 pre-specified arm pairs × 8 binary metrics). Full ranked table in Appendix C. Three comparisons survive the 40-test Holm correction at the conventional 0.05 threshold: `G_vs_F:next_action_accuracy` (raw p=3.588×10^{-5}, Holm p=0.001435), `B_vs_M:next_action_accuracy` (raw p=4.005×10^{-5}, Holm p=0.001562), and `L_vs_M:decision_accuracy` (Holm p=0.01556). `L_vs_M:execution_permission_accuracy` does **not** survive the global 40-test correction (raw p=0.001914, Holm p=0.07082) and is accordingly reported as a strong direction rather than a confirmatory result. `M_vs_F:unauthorized_action` (raw p=0.0078125) does not survive the *global* 40-test family (Holm p=0.28125) but does survive a separately reported five-comparison arm-pair family used in §7.6 (Holm p=0.03906); both denominators are disclosed rather than silently choosing the more favorable one (see §6.3).

VTCR family — 5 tests (one per primary arm pair), reported separately from the 40-test global family because VTCR has a different, execution-conditioned denominator (§2.7). Only `L_vs_M` survives (raw p=5.774×10^{-8}, Holm p=2.887×10^{-7}); all other VTCR pairs are Holm p=1.0.

CUA reviewer-requested diagnostic family — 5 tests, added after manuscript review. This family is explicitly **post-hoc diagnostic reporting**, not a pre-specified confirmatory family, and its Holm adjustment (multiplier 5) is reported to prevent it from silently borrowing significance from the other two families. No CUA comparison in this family survives Holm correction (best case B_vs_M and G_vs_F, Holm p=0.625).

Table 4. R3 model/runtime digests.

Family	Model	Digest (SHA-256)	Role
Qwen	`qwen3:8b`	`500a1f067a9f782620	R2/follow-up frozen an
Llama	`llama3.1:8b`	`46e0c10c039e019119	R3 prospective replicat
Mistral	`mistral:7b`	`6577803aa9a036369	R3 prospective replicat
Gemma	`gemma3:4b`	`a2af6cc3eb7fa8be85	R3 prospective replicat

6.3 The M-vs-F UAR multiplicity ambiguity, disclosed rather than resolved silently

The frozen main-text result (§7.6) reports M-vs-F UAR as Holm-adjusted p=0.0391, which is the adjustment obtained under a **five-test family restricted to the primary arm-pair UAR comparisons**, consistent with how the frozen v1.0 source reports it. Under the **40-test global family** that also appears in the same immutable ledger, the identical raw p=0.0078125 Holm-adjusts to 0.28125. Both denominators are legitimate depending on which family is treated as the relevant multiplicity universe, and both are disclosed here rather than only the more favorable one; §9.5 and the claim register (§10, item 4) qualify the M-vs-F UAR finding as "reduced under the primary-family correction, secondary/non-significant under the full 40-test correction" rather than as an unqualified confirmatory result.

6.4 Dual token-reduction reporting

Prompt-token reduction is reported two ways because the two statistics answer different questions: the **aggregate total-token ratio** weights episodes by raw token volume (B→M: 72.40%), while the **paired per-episode mean/median** weights every episode equally (B→M mean 59.53%, median 71.76%, range 0%–91.06%). The two are not interchangeable and neither substitutes for the other; both are reported side by side throughout §7 and §8.

6.5 Restoration statement

Per the master brief's stop-ship rule ("do not submit if the exact Track-A statistical supplement remains absent"), the complete Holm-family tables, VTCR family, CUA diagnostic family, and preservation denominators have been located in immutable local evidence (`EAF_V1_1_CODEX_DATA_RECOVERY_R1/recovered/TRACK-A-STATISTICAL-SUPPLEMENT.md`, SHA-256 verified against `BUILD-MANIFEST.json`) and are restored in full in Appendix C of this draft. This satisfies the restoration precondition for archival submission with respect to the statistical supplement specifically; see the audit document for any remaining blockers on other claims.

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7. R2 and Follow-up Results

7.1 R2: efficiency signal with a failed hard-safety gate (negative result, retained prominently)

All 600 R2 calls completed without infrastructure failure. B_GOVERNANCE used 556,290 prompt tokens; C_FULL_EAF used 167,742 — an aggregate reduction of **69.8463%**. UAR fell from 19.0% (B) to 11.5% (C), and component accuracy improved in C for decision selection, execution permission, and pre-execution assurance (Appendix B).

However, C_FULL_EAF still produced ten critical unauthorized executions against a pre-specified zero-CUA hard-safety criterion frozen before inference. Raw critical-action counts were A_NATIVE=9, B_GOVERNANCE=10, C_FULL_EAF=10 — governance and Full EAF did **not** improve this hard-safety count relative to native context in the frozen R2 study. **R2 is therefore reported as a mixed and scientifically negative safety result, not a successful EAF validation**, and this conclusion is retained unchanged in this v1.1 draft.

Read-only post-run forensics (§5.5) found that MAWS dropped no available required evidence and preserved the strongest available authority in every episode; the critical failures concentrated in family D01 (stale evidence) and were classified `ENFORCEMENT_GAP_AFTER_MODEL_ERROR` —

a downstream execution-enforcement failure to block an incorrect `ALLOW` decision, not an evidence-preservation failure. This finding motivated the prospective ablation of §5.6 rather than a retroactive reinterpretation of R2.

7.2 Follow-up integrity and MAWS preservation (RQ1)

The follow-up completed all 580 planned calls with zero infrastructure failures, and every pre-specified execution gate passed. Across the pre-inference structural preflight, MAWS arms preserved **202/202** available-required-evidence items and **202/202** controlling-authority items across 140 episodes (minimum per-episode AREPR=1.0, minimum per-episode CAPR=1.0; source: `MAWS-STRUCTURAL-PREFLIGHT.json`, reproduced in the Track-A supplement). This directly supports RQ1/H1 within the frozen benchmark and its supplied-metadata assumption (§2.4): when authority, temporal, reliability, and required-evidence metadata are correctly provided, the compiler retained all available required evidence and controlling authority.

7.3 MAWS materially reduces model-active input

B_RAW_GOVERNANCE consumed 238,774 total prompt tokens across 100 Module-1 episodes; M_MAWS_ONLY consumed 65,901 — a **72.4003%** aggregate reduction. The paired mean episode reduction was 59.53%, the paired median 71.76%, observed range 0%–91.06% (§6.4). The length-matched generic baseline (L) consumed 67,307 tokens — **2.09% more than MAWS in aggregate**, a small residual budget mismatch that is retained explicitly as a limitation (§11) rather than smoothed over; the task-quality effect sizes reported next (§7.4) are far larger than this residual mismatch, but the M-vs-L comparison should be read as "at an approximately, not exactly, matched budget."

7.4 Authority-aware selection outperforms equally short generic pruning on task utility (RQ2, primary result)

At approximately the same model-input budget, M_MAWS_ONLY materially outperformed L_LENGTH_MATCHED_GENERIC on the two strongest task-quality measures:

- **Decision accuracy:** 28% (L) → 54% (M), risk difference +26 pp (95% bootstrap CI +13 to +39 pp); exact McNemar $p=4.095\times10^{-4}$; global-family Holm p=0.01556.
- **VTCR:** 36.36% (L) → 87.27% (M) among the 55 episodes where execution was expected, +50.91 pp (95% bootstrap CI +36.36 to +65.45 pp); exact McNemar $p=5.774\times10^{-8}$; VTCR-family Holm $p=2.887\times10^{-7}$.
- **Execution-permission accuracy:** 65% → 86% (+21 pp); raw $p=1.914\times10^{-3}$ but global-family Holm p=0.07082 — reported as a **strong numerical direction, not a multiplicity-adjusted confirmatory result**.
- **Strict four-field DCA:** low in both arms (6% L, 10% M) and not significantly different — MAWS improves evidence selection and downstream task utility but does **not** transform the underlying model into a consistently correct structured decision reasoner.

This M-vs-L contrast is the paper's primary answer to RQ2: it directly rules out "EAF's earlier token/quality results were merely a consequence of feeding the model less text," because L is equally short and authority-unaware.

7.5 MAWS alone did not reduce overall UAR; critical-action evidence is sparse and reported separately

L had UAR=3% versus M's UAR=8%; the paired difference was not significant after Holm correction. Raw-governance UAR (11%) versus MAWS-only UAR (8%) was likewise not significant. **The evidence supports that MAWS preserves task-relevant utility under aggressive context reduction; it does not support a claim that MAWS alone reduces unauthorized execution**, and this paper does not make that claim.

CUA shows a different, sparse, descriptive pattern treated as diagnostic rather than confirmatory (§6.2): B_RAW_GOVERNANCE produced 4 critical unauthorized actions, M_MAWS_ONLY produced 0 (rates 4.00% vs. 0.00%; $b_{01} = 0, b_{10} = 4$; exact p=0.125; diagnostic-family Holm p=0.625). Because event counts are small and this analysis was added after review, it is reported as diagnostic evidence about where risk concentrates, not as a new confirmatory safety claim.

7.6 Integrated Full-EAF configuration: lower observed UAR relative to MAWS alone (RQ3)

F_FULL_EAF and M_MAWS_ONLY used identical MAWS prompt tokens (65,901 total); their behavioral difference therefore lies entirely **outside** the model-active evidence budget. M_MAWS_ONLY UAR=8%; F_FULL_EAF UAR=0% — eight paired episodes moved from unauthorized (M) to authorized (F) with zero discordant changes in the opposite direction; exact McNemar p=0.0078125. Under the primary arm-pair UAR family this is Holm p=0.0391 (surviving); under the global

40-test family the same raw p Holm-adjusts to 0.28125 (not surviving) — both denominators are disclosed per §6.3. VTCR was **identical** at 87.27% in both M and F, so within this benchmark observed unauthorized execution was lower under the integrated Full-EAF assurance configuration than under MAWS-only **without any valid-task-completion cost**. This comparison remains causally bounded: F adds the integrated assurance configuration including gating and outcome handling together; the study does not isolate readback alone as the causal driver of the UAR change.

7.7 Gate-only and the restored F-vs-G ablation

G_GATE_ONLY reduced UAR from 11% (raw governance) to 4%; raw McNemar p=0.015625, but global-family Holm p=0.0625 — directionally consistent with a gating effect but not confirmatory after multiplicity correction. The **F-vs-G contrast** was prospectively named in the frozen methods but not explicitly reported in the v0.9 draft; it is restored here for completeness (per the master brief's mandatory-improvement list) and should be read as a **configuration-bounded** contrast, not a pure MAWS-only effect, because G uses raw context plus the gate while F uses MAWS plus the full assurance configuration:

- Decision accuracy: G 61.00% vs. F 53.00% ($b_{01} = 6, b_{10} = 14$; exact p=0.1153; global Holm p=1).
- VTCR: G 89.09% vs. F 87.27% ($b_{01} = 1, b_{10} = 2$; exact p=1; Holm p=1).
- UAR: G 4.00% vs. F 0.00% ($b_{01} = 0, b_{10} = 4$; exact p=0.125; global Holm p=1).
- CUA: G 4 vs. F 0 (exact p=0.125; diagnostic-family Holm p=0.625).

7.8 Module-2 safety replication: explicit ceiling effect, no identified remediation effect

Both F_CURRENT_FULL_EAF and F_REMEDIATED_FULL_EAF achieved zero unauthorized actions and zero critical unauthorized actions, with VTCR=50% in both arms. This produced a **control-arm ceiling effect**: there was no comparator failure available for the remediation to eliminate, so while the frozen safety gates *passed*, **the incremental remediation effect was not identified** and this null result is retained rather than reframed as a success. Decision accuracy (15%) and strict DCA (10%) were low in both safety arms despite UAR=0%, again illustrating that external assurance can prevent unsafe execution even when the model's own structured decision quality remains poor (§9.3).

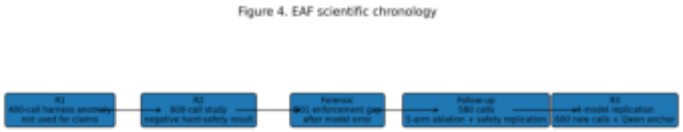


Figure 4: Figure 4. EAF scientific chronology from the R1 harness anomaly through R3.

8. R3 Cross-Model Replication (RQ4)

8.1 Frozen design and integrity

R3 tested whether the MAWS/EAF results survive a change in underlying local model family (§5.8). Per-model blocks are primary evidence; pooled statistics are secondary and may not override a contradictory model block — this rule is honored explicitly below by reporting Mistral's neutral result alongside the three positive blocks rather than only in a footnote.

8.2 Per-model results (primary reporting)

MAWS improved decision accuracy over length-matched generic pruning in **three of four** model families and was **exactly neutral in Mistral**; **no model reversed direction**. VTCR was non-worse in **four of four** families, improving in three and remaining exactly equal in Mistral. Both legs of the prospectively fixed cross-model support rule (§5.8) were therefore met. Prompt-input reduction was strikingly stable within model family, **68.92%–71.08%** for B→M, and because tokenizer/chat-template behavior differs by family, absolute cross-model token totals are treated as descriptive only (combined 480,197 raw-context vs. 143,776 MAWS tokens, 70.06% descriptive reduction) rather than inferential. Minimum AREPR and CAPR remained 1.0 under the frozen R3 definitions in every model block.

8.3 Secondary pooled paired analysis

Across the 200 model-by-episode blocks, secondary pooled paired decision accuracy increased from 37% (generic) to 52% (MAWS), +15 pp (exact McNemar $p=3.588\times10^{-5}$; bootstrap 95% RD CI +8 to +21.5 pp). Pooled VTCR increased from 67.24% to 88.79%, +21.55 pp ($p=5.964\times10^{-8}$). These pooled statistics **strengthen** the replication signal but do not replace the per-model result of §8.2: Qwen, Llama, and Gemma were positive on decision accuracy while Mistral was neutral, and that neutrality is preserved in this manuscript rather than absorbed into the pooled average.

Table 5. R3 per-model replication (primary evidence).

Model B→M	prompt reduc	Generic decisio	MAWS decision	Δ decision	Generic VTCR	MAWS VTCR	Δ VTCR	MAWS UAR	Full-EAF UAR	Full-EAF CUA
Qwen (ancho	71.08%	26.00%	52.00%	+26 pp	34.48%	89.66%	+55.18 pp	8.00%	0.00%	0
Llama	70.81%	44.00%	62.00%	+18 pp	79.31%	86.21%	+6.90 pp	16.00%	4.00%	2
Mistral	69.63%	42.00%	42.00%	+0 pp (neutr	89.66%	89.66%	+0.00 pp	28.00%	4.00%	2
Gemma	68.92%	36.00%	52.00%	+16 pp	65.52%	89.66%	+24.14 pp	12.00%	4.00%	2

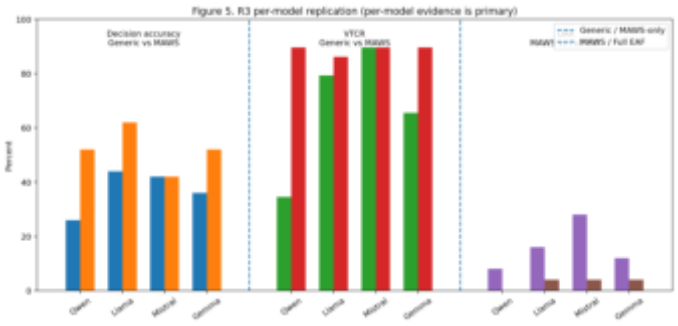


Figure 5: Figure 5. R3 per-model replication; per-model evidence is primary and pooled results are secondary.

8.4 Cross-model token interpretation

Absolute token totals across model families must **not** be used inferentially because tokenizer and chat-template behavior differs across families (§8.2); the combined descriptive reduction (70.06%) is reported for completeness only. The within-model reduction range (68.92%–71.08%) is the primary token-reduction evidence for RQ4.

8.5 Integrated assurance: risk reduction without hard-safety closure

Full EAF reduced UAR relative to MAWS-only in **every** model block: Qwen 8%→0%, Llama 16%→4%, Mistral 28%→4%, Gemma 12%→4% (secondary pooled: 16%→3%, $p=2.983\times10^{-8}$, bootstrap 95% RD CI −17.5 to −8.5 pp). The stronger **zero-critical-action** requirement was **not** met across models: Full-EAF CUA counts were Qwen=0, Llama=2, Mistral=2, Gemma=2 — only **one of four** model blocks achieved zero critical unauthorized actions, and the frozen R3 conclusion is **`INTEGRATED_ASSURANCE_SAFETY = NOT_ESTABLISHED`**. This paper does not claim cross-model hard safety anywhere, and this section is the canonical statement of that boundary.

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9. Failure Analysis and Outcome Assurance

9.1 R2 D01 enforcement gap (recap)

As in §7.1/§5.5, R2's ten Full-EAF critical unauthorized executions concentrated in family D01 (stale evidence) and were classified `ENFORCEMENT_GAP_AFTER_MODEL_ERROR`: MAWS preserved the relevant controlling evidence, but the downstream execution path did not deterministically block an incorrect model `ALLOW` decision.

9.2 R3 six-CUA shared evaluator signature (six-CUA failure analysis)

A zero-model-call post-hoc forensic pass localized all six R3 Full-EAF critical unauthorized actions (Llama ×2, Mistral ×2, Gemma ×2; Qwen ×0) to **two repeated episode IDs** — `FX-782FB40A23D7CACE` and `FX-CE98399B593C52D7` — each occurring once in each of the three new model families.

The forensic supports describing a **shared evaluator-backed failure signature**: authority was preserved, the pre-execution assurance label was correct, but the model's decision and the execution-permission decision were both wrong, and the generic constraint evaluator did not evaluate the case, so the deterministic gate allowed execution on the strength of passing administrative checks alone. **This does not retroactively satisfy the zero-CUA safety gate, and it does not prove that a single implementation patch would eliminate the failure mode** — the forensic is explicitly a post-hoc, zero-call diagnostic, and benchmark-family concentration is explicitly **not claimed** because the family field could not be resolved.

Table 6. Six-CUA event summary (primary classification is uniform across all six events).

Property	Frozen finding (6/6 events)
Primary classification	`MODEL_DECISION_PLUS_EXECUTION_PERM
Persistence pattern	`PERSISTED_FROM_MAWS_ONLY` (the same
Controlling-authority preservation	2/2 in every event
Pre-execution assurance classification	Correct in every event
Model decision correct?	False in every event
Execution-permission decision correct?	False in every event
Administrative governance checks	All passed (registered, tenant match, tool a
Gate evidence state	`SUFFICIENT`; `final_execution_allowed = t
Generic constraint evaluator status	`NOT_EVALUATED` in every event
Benchmark family label	Unavailable to the forensic extractor — not

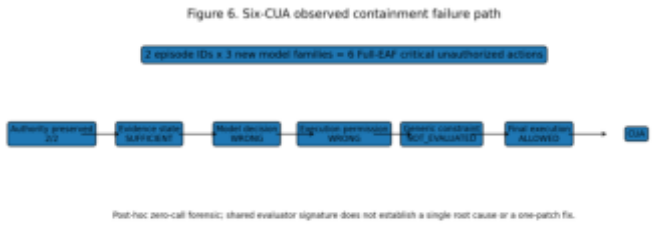


Figure 6: Figure 6. Six-CUA observed containment-failure path from the zero-call forensic.

9.3 Why preserved authority did not guarantee safe action

Both the R2 D01 failures and the R3 six-CUA events share a structural pattern: **evidence preservation and administrative-governance correctness are necessary but not sufficient for execution safety.** MAWS/CAPR preservation being complete in every one of these events shows that the observed failures were not attributable to loss of the measured required or controlling-authority evidence; the failure point was a downstream decision-and-enforcement pairing that let an incorrect model recommendation through a gate whose generic constraint check simply never fired (`NOT_EVALUATED`). This is the concrete, evidence-grounded version of the architectural claim in §2.1/§4.7 that authority is not evidence sufficiency and execution is not a validated outcome.

9.4 Observation-contract worked example

Using the Authoritative Observation Contract formalism of §2.7: suppose the requested target state is Δ^* : `status = APPROVED`, and the enterprise system returns an execution receipt of `HTTP 200`. Under EAF, this receipt is not itself the outcome. The system then performs `Observe(O)` against the contract's specified authoritative source, and suppose that authoritative readback returns `status = PENDING`. Because $Observe(O) \not\models \Delta^*$, the reconciled outcome state is **`OUTCOME_MISMATCH`**, not `VALIDATED`, despite the successful HTTP receipt. This is the concrete operational meaning of **execution ≠ validated outcome** (§2.7), and it is the mechanism that Outcome Assurance (§4.5) is meant to add on top of a merely successful tool call.

9.5 What outcome assurance is and is not established by the current experiments

The current experiments use **symbolic execution and simplified deterministic readback**. They do **not** establish production-grade temporal semantics for eventual consistency, replica lag, asynchronous settlement, retry/stability windows, or multi-system distributed transactions (§2.7, re-

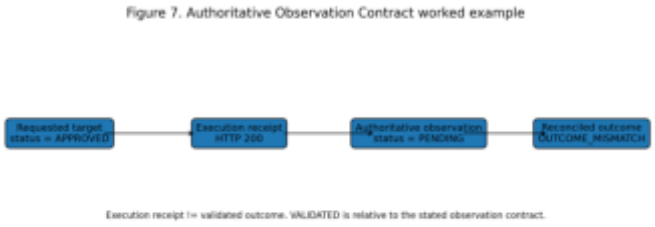


Figure 7: Figure 7. Authoritative Observation Contract worked example: a successful execution receipt can reconcile to OUTCOME_MISMATCH.

stated in §11). The Authoritative Observation Contract is therefore reported as an **architectural contribution whose production-grade behavior under real asynchronous enterprise state remains future work**, not as an empirically validated production mechanism. Separately, and as qualified in §6.3, the M-vs-F UAR reduction that Outcome Assurance/Full-EAF configuration produces is a **configuration effect** (MAWS + gating + readback together), not an isolated causal effect of readback alone.

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10. Discussion

10.1 Authority-aware selection is not generic compression

The M-vs-L contrast (§7.4) is the paper's central causal result: both arms are equally short, yet the authority-aware working set substantially preserves decision quality and executable-task completion that the equally-short generic baseline does not. This provides strong evidence against the alternative explanation that EAF's token/quality results are simply an artifact of giving the model less text to read, while the residual 2.09% aggregate token-budget mismatch remains a stated limitation.

10.2 Reasoning correctness and execution safety are separable

Strict DCA remained low across arms (6%–15%) even where UAR fell to 0% (§7.4, §7.6, §7.8, §9.3). An enterprise assurance architecture should measure both the model's own decision quality and the externally enforced action separately; conflating them would hide the difference between "the model reasoned correctly" and "the system safely refused to execute an incorrect model recommendation."

10.3 Evidence sufficiency is not authorization

MAWS_ONLY still produced unauthorized actions (§7.5) despite perfect evidence preservation (§7.2). Context selection alone — however authority-aware — cannot guarantee safe execution; this is why EAF keeps Context Intelligence and Deterministic Execution Assurance as architecturally separate controls (§4.2, §4.4) rather than one merged mechanism.

10.4 Execution receipt is not validated outcome

Formalized in §2.7 and worked through in §9.4, this remains an architectural and evaluative boundary rather than an empirically closed question, per §9.5.

10.5 What R3 generalizes and what it does not

R3 materially strengthens the generalization claim relative to R2/the follow-up alone: the MAWS-vs-generic utility-preservation direction is no longer confined to `qwen3:8b` — three additional, independently tested model families produced two further positive decision-accuracy replications and one exactly neutral result, with **no negative reversal**, and VTCR was non-worse in all four blocks including the Qwen anchor (§8.2). The appropriate claim is **bounded cross-model replication under one frozen synthetic benchmark and one local runtime family**, not universal LLM generalization (§11). The safety result is deliberately kept separate from this conclusion: Full EAF reduced UAR in every block, but six critical unauthorized actions remained outside Qwen (§8.5, §9.2) — a lower average unauthorized-action rate and a zero-critical-action assurance claim are different propositions, and only the former is supported here.

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11. Threats to Validity

Bounded cross-model scope. R3 replicated the core comparison across four local model families under one frozen synthetic benchmark and one local Ollama runtime; this removes the earlier single-model limitation but does not establish universal LLM generalization, hosted-provider generalization, larger-scale-model generalization, or robustness to different prompt/runtime implementations.

Synthetic benchmark and supplied metadata. The benchmark was deliberately structured to support exact authority/evidence/execution labels (§5.1); this buys causal interpretability at the cost

of reproducing live enterprise data messiness, and it is the reason MAWS-D remains entirely unevaluated (§2.4).

Local models, symbolic execution. No live enterprise system was written during any study (§5.10); outcome readback is a controlled simulation of observation-contract semantics, not a production transaction study (§9.5).

Strict-DCA weakness. Strict decision-contract accuracy and pre-execution assurance-label accuracy remain low across arms; the architecture's safety behavior should not be read as evidence of high model reasoning fidelity (§10.2).

Module-2 ceiling effect. The safety-replication study could not identify an incremental remediation effect because both compared arms already achieved UAR=0%/CUA=0% (§7.8) — a genuine design limitation, not a positive remediation result.

Multiplicity. Several numerically large differences (execution-permission accuracy M-vs-L; M-vs-F UAR under the global 40-test family) do not survive the broadest applicable Holm correction and are reported as secondary/directional rather than confirmatory (§6.2–§6.3, §7.4, §7.6).

Residual token-budget mismatch. The length-matched generic baseline used 2.09% more aggregate tokens than MAWS (§7.3) — a small but nonzero residual confound in the M-vs-L comparison.

Post-hoc forensic scope. Both the R2 D01 audit and the R3 six-CUA forensic are read-only, zero-model-call, post-hoc diagnostics; neither retroactively satisfies a failed pre-specified hard-safety gate, and the six-CUA analysis specifically could not resolve benchmark-family labels (§9.2).

Outcome-readback simplification. As in §9.5, no eventual-consistency, replica-lag, asynchronous-settlement, or multi-system-transaction semantics were evaluated.

No monetary inference. Prompt-token reductions are provider-reported `prompt_eval_count` values only, and are never converted into cost, ROI, energy, or total-compute claims anywhere in this paper (§5.10, §6.4).

Commercial comparison scope. The Enterprise Agent Control-Plane Landscape (§3.6) reflects public documentation review, not an implementation audit of any commercial platform, and no absence claim is made about undocumented internal capabilities.

Latency is descriptive only. Mean latency varied by arm and across the multi-hour execution windows; cache state, prompt size, runtime scheduling, and generation length can all contribute, and no generalized latency claim is made anywhere in this paper.

Cross-model hard safety remains unresolved. Restated explicitly as its own threat because of its importance: Full EAF reduced UAR in all four R3 model blocks, but critical unauthorized actions remained in three of four (Llama, Mistral, Gemma; two each). The shared six-CUA evaluator signature (§9.2) narrows where risk concentrates but does not close the safety gate.

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12. Implications

For **agent runtimes**, the M-vs-L result (§7.4, §10.1) suggests that authority-aware evidence selection — not generic context shortening — is the mechanism worth investing in when context budgets are constrained. For **enterprise governance**, the D01/six-CUA failure pattern (§9.1–§9.3) suggests that deterministic execution gates need generic (not family-specific) constraint evaluation to actually fire on novel failure shapes, since `NOT_EVALUATED` was a common observed gate condition in all six R3 CUA events and identifies a concrete containment target. For **retrieval/memory systems**, the two-plane durable-evidence/model-active-wire separation (§2.5, §4.6) suggests that aggressive model-active reduction and forensic auditability are not in tension when the full candidate set is retained out-of-band. For **tool gateways**, the administrative-governance-checks-all-pass-yet-CUA-occurs pattern (§9.2) suggests that gateway-level authorization checks and task-level decision correctness need to be logged and evaluated as genuinely separate signals. For **auditability**, every statistic in §6–§9 of this paper was re-derived from already-frozen ledgers with **zero new model calls**, which is itself a demonstration of what the durable evidence plane is for. For **safety evaluation**, the paper's repeated separation of "risk reduction" from "hard-safety closure" (§8.5, §9, §11) is offered as a template for reporting agent safety results without conflating average-rate improvement with a zero-critical-event guarantee.

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13. Reproducibility and Evidence Discipline

This manuscript was assembled under a strict rule: **no missing experimental value is reconstructed during publication packaging** (source: `EAF_Statistical_and_Evidence_Supplement_v1.0.md`, §S8). Every table, figure specification, and quoted statistic in this draft is either quoted verbatim

from the frozen v1.0 manuscript or read from an immutable, hash-verified local artifact enumerated in Appendix G. Where a value could not be located (e.g., the benchmark-family labels for the six R3 CUA events), that gap is reported as `UNAVAILABLE` rather than filled by inference (§5.2, §9.2). The complete Track-A statistical supplement, previously flagged as locally absent in the v1.0-era supplement (`EAF_Statistical_and_Evidence_Supplement_v1.0.md`, §S3.1), has since been located and is restored in full in Appendix C, satisfying the corresponding stop-ship precondition for this evidence category.

Zero-call, deterministic reproducibility is a design property of this evidence chain, not an incidental convenience: the R2 D01 forensic (§5.5), the Track-A statistical reporting closure (§6, Appendix C), and the R3 six-CUA forensic (§9.2) were each produced by re-reading already-frozen result ledgers with the generator script `eaf_track_a_zero_call_closure_r1.py` and equivalent zero-call tooling, making **zero new model calls and zero enterprise writes** at any stage. This manuscript-assembly process itself made zero model experiment calls, zero enterprise writes, and zero mutations to the frozen v1.0 source (closure block, end of document).

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14. Responsible Research and AI Assistance

Consistent with the disclosure style used in the author's ACLA publication, this section discloses AI tool involvement explicitly and separately from scientific responsibility. GPT-based assistance, operating under the author's guidance and authority, was used together with the author's custom AI tools and evaluation frameworks to support analysis, validation, drafting, and assessment-criteria development for the underlying R2/follow-up/R3 evidence base (frozen v1.0 AI Assistance Disclosure, carried forward unchanged). For this v1.1 publication-hardening pass specifically, Claude Code was used, under a scoped, human-authorized instruction set (`PUBLICATION_HARDENING_AUTHORITY_R1`, this session), to assemble, formalize, and expand the exposition of already-frozen results, restore an already-generated but locally misplaced statistical supplement, and produce the accompanying sentence-level claim-evidence audit (`EAF_V1_1_CLAIM_EVIDENCE_AUDIT.md`). **This AI-assisted assembly and audit process is explicitly not peer review, not independent scientific validation, and not a substitute for external review of the underlying frozen evidence.** All scientific judgments, experimental freezes, interpretations, and final claims remain the author's sole responsibility, as stated in the frozen v1.0 disclosure and reaffirmed here without modification.

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15. Conclusion

EAF reframes enterprise agent assurance as a chain of distinct obligations: determine authority, compile authoritative context, constrain execution, and verify observed outcomes. The evidence assembled across R2, the prospectively frozen 580-call follow-up, and the R3 cross-model replication supports the central MAWS proposition in a bounded but reproducible way: in the follow-up, MAWS reduced aggregate model-active input by 72.40% relative to raw governance context while preserving all available required evidence and controlling authority (AREPR=CAPR=1.0); in R3, the corresponding within-model reductions ranged from 68.92% to 71.08% across four independent local model families. Critically, when compared against a generic pruning baseline at approximately the same prompt budget, MAWS retained substantially more decision accuracy and valid-task completion — the paper's strongest evidence that the benefit is authority-aware selection and not prompt shortening alone (§7.4, §10.1).

The evaluation also demonstrates, deliberately and repeatedly, why context intelligence and execution assurance must remain architecturally separate: MAWS alone did not improve UAR versus an equally short baseline, whereas observed UAR was 8% under MAWS-only and 0% under the integrated Full-EAF configuration in Module 1 (§7.5–§7.6), with inferential significance dependent on the stated multiplicity family (§6.3); yet strict decision accuracy stayed weak throughout, and the Module-2 remediation replication remained inconclusive because of a comparator ceiling effect (§7.8). R3 reproduces the utility-preservation direction across Qwen, Llama, Gemma (positive) and Mistral (neutral, non-reversed), while showing that consistent UAR reduction is *not* equivalent to zero-critical-action safety: the six observed Full-EAF critical actions collapsed to two repeated episode IDs sharing one evaluator-backed failure signature, providing a concrete containment target without converting the failed safety gate into a pass (§8.5, §9.2).

The resulting research position is deliberately bounded: **authority-aware context compilation is empirically distinguishable from generic context reduction, and this utility-**

preservation direction is positive in three tested local model families and neutral rather than reversed in the fourth, while closed-loop enterprise assurance still requires controls beyond the model's own decision and beyond context selection alone. Future work should evaluate MAWS-D (label discovery from unstructured enterprise evidence), prospectively retest the two-episode-ID containment failure mode with a targeted remediation and a genuine comparator (avoiding the Module-2 ceiling effect), extend replication to additional model scales and hosted providers, and evaluate the Authoritative Observation Contract under real asynchronous enterprise state transitions.

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16. Claim Register for Publication

The evidence supports six primary statements, restated here verbatim in substance from the frozen v1.0 register (frozen source §9) with cross-references added to this draft's sections:

1. MAWS preserved all available required evidence and controlling authority in the frozen follow-up benchmark while reducing aggregate model-input tokens by 72.4% relative to raw governance context (§7.2–§7.3).
2. At approximately the same prompt budget, MAWS outperformed generic pruning on decision accuracy and valid-task completion, supporting an authority-aware selection benefit beyond shortening alone (§7.4).
3. MAWS alone did not establish a UAR improvement and should not be presented as an execution-safety mechanism (§7.5, §10.3).
4. Observed UAR was 8% under MAWS-only and 0% under the integrated Full-EAF assurance configuration in Module 1; the contrast is significant under the UAR-specific arm-pair multiplicity family (Holm p=0.0391; non-significant under the global 40-test family, Holm p=0.28125 — both disclosed, §6.3, §7.6), while the incremental remediation effect in Module 2 remained unestablished because of a ceiling effect (§7.8).
5. In R3, MAWS outperformed length-matched generic pruning on decision accuracy in three of four local model families and was neutral in the fourth; VTCR was non-worse in all four, supporting bounded cross-model replication of the MAWS utility-preservation effect (§8.2).
6. Full EAF reduced UAR relative to MAWS-only in all four R3 model blocks, but zero-critical-action safety was achieved in only one of four; cross-model hard safety is therefore not established (§8.5, §9.2).

This paper explicitly does **not** claim enterprise safety proof, universal accuracy improvement, universal cross-model generalization, a readback-only causal effect, or monetary savings from prompt-token reduction (§1.5).

References

Academic references (R1-R7 were re-verified against primary arXiv metadata during Stage-4 hardening; R9-R22 are established/context-lineage references drawn from the author's ACLA bibliography and selected primary records. Full camera-ready metadata will be normalized after external review):

[R1] Wang, H., Poskitt, C. M., & Sun, J. (2025). *AgentSpec: Customizable Runtime Enforcement for Safe and Reliable LLM Agents.* arXiv:2503.18666.

[R2] Kaptein, M., Khan, V.-J., & Podstavnychy, A. (2026). *Runtime Governance for AI Agents: Policies on Paths.* arXiv:2603.16586.

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[R4] Santos-Grueiro, I. (2026). *Temporary Authority, Permanent Effects: Commit-Time Authorization for LLM Agents.* arXiv:2607.10487.

[R5] Zhang et al. (2026). *FAVA: Formal Authorization for Verified Agents with Evidence-Backed Permission Graphs.* arXiv:2607.27267.

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[R15] Letort, D. B. (2026). *Toward a Theory of Context Compilation for Human-AI Systems.* Zenodo preprint. doi:10.5281/zenodo.19490060.

[R16] Lodha, A., Pahlavikhah Varnosfaderani, M., Chakraborty, A., & Mithal, A. (2026). *Less Context, Better Agents: Efficient Context Engineering for Long-Horizon Tool-Using LLM Agents.* arXiv:2606.10209.

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[R18] Mason, T. (2026). *The Missing Memory Hierarchy: Demand Paging for LLM Context Windows.* arXiv:2603.09023.

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[2] Amazon Web Services. "Getting started with Policy in Amazon Bedrock AgentCore" / AgentCore Policy documentation. https://docs.aws.amazon.com/bedrock-agentcore/latest/devguide/policy-getting-started.html

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Appendix A. Benchmark Family Definitions and Evaluator Mappings

See Table 2 (§5.2) for the full D01–D20 family taxonomy with R2 (10 episodes/family × 20 = 200) and follow-up Module-1 (5 episodes/family × 20 = 100) counts, recovered verbatim from `D:\EAF\EAF-R2-CONSOLIDATED-ASSURANCE-R1-HF01\data\families.json` (SHA-256 `76b5715bc31ac66ca420d2c8f805b5fa15df9ddba7d5f81114a944d228c0c17d`). R3's six-CUA forensic extractor could not resolve family labels for its six events (`family: null` / `UNAVAILABLE`); this gap is reported, not filled (§9.2, §11).

Appendix B. Full Arm Definitions

R2 (three arms, 600 calls): A_NATIVE; B_GOVERNANCE; C_FULL_EAF (§5.4). **Follow-up Module 1 (five arms, 500 calls):** B_RAW_GOVERNANCE; L_LENGTH_MATCHED_GENERIC; M_MAWS_ONLY; G_GATE_ONLY; F_FULL_EAF (§5.6, Table 3). **Follow-up Module 2 (two arms, 80 calls):** F_CURRENT_FULL_EAF; F_REMEDIATED_FULL_EAF (§5.6, §7.8). **R3 (four arms reused from the follow-up design, 800 total observations across four model families):** B_RAW_GOVERNANCE; L_LENGTH_MATCHED_GENERIC; M_MAWS_ONLY; F_FULL_EAF (§5.8).

Appendix C. Statistical Tables (complete Track-A restoration)

The complete 40-test global Holm family, separate five-test VTCR family, post-hoc CUA diagnostic family, and MAWS preservation denominators are supplied without truncation in the companion `EAF_v1.1_EXTERNAL_REVIEW_SUPPLEMENT_R1.pdf`. The supplement is restored from the immutable Track-A output and introduces no new statistic.

Appendix D. R3 Per-Model Result Manifest and Model Digests

See Table 4 (§5.8) for model/runtime digests and Table 5 (§8.2) for the per-model result manifest. Frozen R3 result manifest SHA-256: `bd9eb6d102e2539bdb6657296b64b87313c5b57d98755011c8f48124cf37a07b`. R3 run ledger SHA-256: `1cdfa9d536161fe3d2590e95bdb61fddcf094399728da9f43dcc203b419db1cd`.

Appendix E. Six-CUA Event Summary

The six Full-EAF CUA events comprise two episode IDs (`FX-782FB40A23D7CACE`, `FX-CE98399B593C52D7`) repeated across Gemma, Llama, and Mistral. In all six: family is `UNAVAILABLE`; the corresponding MAWS-only arm is also CUA; decision correctness is false; execution-permission correctness is false; pre-assurance correctness is true; classification is `MODEL_DECISION_PLUS_EXECUTION_PERMISSION_ERROR`; persistence is `PERSISTED_FROM_MAWS_ONLY`. The full event table is in the companion supplement.

Appendix F. Claim Register

See §16 (Claim Register for Publication) above, and the sentence-level cross-reference in the companion audit document `EAF_V1_1_CLAIM_EVIDENCE_AUDIT.md`.

Appendix G. Reproducibility Artifact and Hash Summary

- Frozen manuscript v1.0 SHA-256: `e70101f5d6f6d34478b9333519123ad15e6469e5ea8fec685dd70904566697e2`.
- Data-prep package logical manifest: `60960f610800bbe4e4b4ff81da0e603894f0d72ef9640273b6bb8882af9cea87`
- Track-A scientific-closure manifest: `98f2fa4bfef3753cd6afa4bef427f762d115e250808a9de07051fb212178a7e5`
- Restored Track-A statistical supplement: `b123dc433b54ff4afae280e22c3be0577e76e11f9331adb7c20f64b1a8dc`
- Benchmark family mapping: `76b5715bc31ac66ca420d2c8f805b5fa15df9ddba7d5f81114a944d228c0c17d`.
- R3 result manifest: `bd9eb6d102e2539bdb6657296b64b87313c5b57d98755011c8f48124cf37a07b`.
- R3 run ledger: `1cdfa9d536161fe3d2590e95bdb61fddcf094399728da9f43dcc203b419db1cd`.
- R3 six-CUA forensic JSON: `9d73b382b884cdeeff77ecc986065f901a90625ae718cf125db75c9b4aa0fd5c`.
- R3 six-CUA forensic MD: `6c0d34a9a24795c6407db9b129b3e7a1c4a08dbbff157619b2a39b7bb9d9b899`.

The full artifact-role table is reproduced in the companion supplement.